

Unifying Special Relativity and Quantum Mechanics

Lecture 32 — The Fermionic Sector, the Generation of Mass, and the Electroweak Phase Transition

(the final lecture)

(the three generations and their representations; the electroweak currents split into electromagnetic, charged and neutral; $e = g_L \sin \theta_W$; muon decay and the measurement of v ; neutron beta decay; why a chiral theory forbids fermion masses and how the Yukawa couplings restore them, including the conjugate doublet $\bar{H} = i\sigma_2 H^*$ for the up quark; the observational summary; and finally the finite-temperature potential and the electroweak phase transition at $T_{EW} \simeq 160 \text{ GeV}$)

Typed version of the handwritten notes, with added explanations in blue boxes

Added explanation (not in the original notes) — How to read this document

Conventions. Exactly as in Lectures 1–31: normal text and the yellow / pink / grey / green boxes are a faithful transcription of the handwritten sheets, and the text colours follow the pens of the original (blue, magenta, green, red, violet, pencil, brown). Everything inside a blue box like this one is material I have added; it is not in the original notes.

Pagination.

1. **The first two sheets are a repeat.** This scan opens at sheet 17, but Lecture 31 already ran [p. 7*****]–[p. 18*****]. Sheets 17 and 18 are duplicated from the end of that scan. Section 1 records their content compactly; for the full transcription see Lecture 31, Sections 11–12.
2. **Everything from [p. 19*****] on is new,** and it is all one chapter — still *A Standard Model primer*, begun on [p. 1*****] in Lecture 30. So every margin marker carries the same *fifteen* stars, [p. 17*****]–[p. 36*****]. No blanks, no gaps, no restarts.

PDF page n is sheet $n + 16$, throughout.

And this time it does *not* stop mid-development. For the first time in many lectures the scan reaches a genuine ending: [p. 36*****] closes with the electroweak phase transition in the early universe, $T_{EW} \simeq 160 \text{ GeV}$ at $t \simeq 10^{-10} \text{ s}$ after the Big Bang. That is the end of the course. Accordingly, the final section of this document is not the usual “where this is going” but a **retrospective** — what the thirty-two lectures built, and what they deliberately left open.

Slips and reading notes, flagged in place.

1. On [p. 22*****] the object expanded in the “we write” step is written $g_L W_\mu^A \sigma^A - g_Y B_\mu$, i.e. *without* the $\frac{1}{2}$ that appears in $\mathcal{L}_{\text{int}}$. The factor is restored on [p. 24*****], where the matrix is multiplied by $g_L/2$. Kept as in the original, with the bookkeeping noted in a blue box.
2. On [p. 27*****] the two braces on the right of the charged-current box are drawn with two similar circled stars. They are distinguished here as ($\otimes$) for the two leptonic lines and ($\odot$) for the quark line, since [p. 27*****] refers to the first for muon decay and [p. 29*****] to the second for beta decay.
3. On [p. 36*****] the finite-temperature potential is written $V(H, T) = (-\mu^2 + cT^2)H^\dagger H + \lambda(H^\dagger H)$; the quartic is missing its exponent and should read $\lambda(H^\dagger H)^2$. Corrected above.
4. On [p. 31*****] the estimate is quoted as $y_e \simeq 10^{-6}$; the precise value is $y_e = \sqrt{2} m_e/v \simeq 2.9 \times 10^{-6}$. Order of magnitude as stated.

5. On [p. 35*****] the labels on the coupling-versus-mass plot are hard to read in the scan. The six couplings measured at the LHC are μ, τ, b, W, Z, t , and the figure is redrawn with those labels.

Notation. g_L the $SU(2)_L$ coupling, g_Y the $U(1)_Y$ coupling, $s_W \equiv \sin \theta_W$, $c_W \equiv \cos \theta_W$, $v^2 = \mu^2/\lambda$, $e = g_L s_W$. Chiral projectors $P_L = \frac{1}{2}(1 - \gamma^5)$, $P_R = \frac{1}{2}(1 + \gamma^5)$. Fermion multiplets $Q_u = (u_L, d_L)^T$, $L_e = (\nu_{eL}, e_L)^T$, with right-handed singlets u_R, d_R, e_R . Labels $(\mathbf{n}_c, \mathbf{n}_L, Y)$ for $(SU(3)_c, SU(2)_L, U(1)_Y)$, and $Q = T^3 + Y$.

Added explanation (not in the original notes) — Where we left off — recap of Lecture 31

Lecture 31 completed the *bosonic* sector:

- The neutral mass matrix $M^2 = \frac{v^2}{4} \begin{pmatrix} g_L^2 & -g_L g_Y \\ -g_L g_Y & g_Y^2 \end{pmatrix}$ has $\det = 0$, so one eigenvalue vanishes. Diagonalising by the Weinberg rotation gives $Z_\mu = c_W W_\mu^3 - s_W B_\mu$ with $M_Z^2 = \frac{v^2}{4}(g_L^2 + g_Y^2)$, and a massless $A_\mu = c_W B_\mu + s_W W_\mu^3$, with $\tan \theta_W = g_Y/g_L$ and $M_Z = M_W/c_W$.
- Charges follow from the unbroken $U(1)$: h and Z neutral, $W^\pm$ of charge ± 1 (because $Q = T^3 + Y = t^3$ on the gauge fields — the Georgi–Glashow situation again).
- The mass ladder $M_W < M_Z < M_h < v$, and the three categories of bosonic interaction: Higgs self-couplings, Higgs–gauge couplings proportional to M_V^2/v^2 , and gauge self-couplings whose complete list follows from charge conservation.

The scan then opened the fermionic sector, defined P_L, P_R , and stopped at the sentence “the left and right components transform differently”. **This lecture is the continuation of that sentence**, and it runs to the end of the course.

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PART I — The Standard Model fermions [p. 17*****]–[p. 21*****]

1 The duplicated sheets

[p. 17*****]–[p. 18*****]

Added explanation (not in the original notes) — These two sheets already appeared at the end of Lecture 31

Sheets [p. 17*****] and [p. 18*****] are physically the same pages as the last two of the Lecture 31 scan. Their content is transcribed in full there (Sections 11–12). Compactly:

Sheet [p. 17***].** Category 3 of the bosonic interactions — gauge boson self-interactions, from $-\frac{1}{4}W_{\mu\nu}^A W^{A,\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu}$ rewritten in the mass eigenstates. Charge conservation alone lists them: the trilinear vertices AW^+W^- and ZW^+W^- , and the quartics AAW^+W^- , ZZW^+W^- .

Sheet [p. 18***].** The remaining quartics ZAW^+W^- and $W^+W^-W^+W^-$; then the section *The fermionic sector* opens with a Dirac fermion split as $\Psi = \Psi_L + \Psi_R = P_L\Psi + P_R\Psi$, $P_L = \frac{1}{2}(1 - \gamma^5)$, $P_R = \frac{1}{2}(1 + \gamma^5)$, and the statement that the Standard Model is a *chiral* theory because the left and right components transform differently under $SU(3)_c \times SU(2)_L \times U(1)_Y$.

The next sheet takes that statement and makes it concrete. Everything from there on is new.

2 The three generations, and one generation's content

[p. 19*****]

The Standard Model fermions are

	1 st generation	2 nd generation	3 rd generation
QUARKS	up (u) down (d)	charm (c) strange (s)	top (t) bottom (b)
LEPTONS	electron (e) electron neutrino (ν_e)	muon (μ) muon neutrino (ν_μ)	tau (τ) tau neutrino (ν_τ)

the mass always increases going from left to right

Let us consider for simplicity **only one generation**.

$u_L, u_R, d_L, d_R, e_L, e_R, \nu_{eL}, \nu_{eR}$

u_L, d_L combined into $SU(2)$ doublet

$$Q_u = \begin{pmatrix} u_L \\ d_L \end{pmatrix} \sim (\mathbf{3}, \mathbf{2}, \frac{1}{6})$$

△ In the **original version** of the Standard Model, we **do not have right-handed neutrinos**

ν_{eL}, e_L combined into $SU(2)$ doublet

$$L_e = \begin{pmatrix} \nu_{eL} \\ e_L \end{pmatrix} \sim (\mathbf{1}, \mathbf{2}, -\frac{1}{2})$$

$$e_R \sim (1, 1, -1)$$

$$u_R \sim (3, 1, \frac{2}{3})$$

$$d_R \sim (3, 1, -\frac{1}{3})$$

[p. 20*****]

Added explanation (not in the original notes) — The structure of the table, and the one field that is missing

Why the left-handed fields are doublets and the right-handed ones singlets. This is *the* defining choice of the Standard Model, and it is not derived from anything — it is read off from experiment. The $W^\pm$ boson turns an electron into a neutrino, and a down quark into an up quark, but *only* the left-handed ones. Putting (ν_L, e_L) into a doublet makes that a gauge interaction; leaving e_R as a singlet makes it inert under $W^\pm$. Parity violation is built in at the level of the field content.

Counting one generation. Fifteen Weyl fermions: Q_u (3 colours $\times$ 2 isospin = 6), u_R (3), d_R (3), L_e (2), e_R (1). Sixteen if one adds ν_R . *This is exactly one 16 of $SO(10)$* — the observation that started grand unification.

The crossed-out ν_{eR} . A right-handed neutrino would be a complete gauge singlet, $(1, 1, 0)$: it feels no force at all. In the *original* Standard Model it is simply not included, and the immediate consequence appears on [p. 31*****]: **the neutrino comes out exactly massless**. Neutrino oscillations, observed from 1998 onwards, prove that at least two neutrinos have mass — so this is the one place where the model as written here is known to be *wrong*. The red warning triangle is well placed.

The mass ordering. That masses rise from generation to generation is an empirical pattern spanning eleven orders of magnitude ($m_\nu \lesssim 0.1$ eV to $m_t = 173$ GeV), with no explanation in the Standard Model. Every one of those masses is an independent free parameter — see [p. 31*****]–[p. 34*****].

3 Hypercharges and electric charges; colour

[p. 20*****]–[p. 21*****]

COMMENTS

1) The hypercharge assignments are such that we reproduce the correct electric charge under $U(1)_{\text{em}}$ unbroken

Example: $L_e \sim (\mathbf{1}, \mathbf{2}, -\frac{1}{2})$

The generator of $U(1)_{\text{em}}$ is $Q = T^3 + Y$

In this case, $T^3 = \frac{\sigma^3}{2}$ since L_e transforms in the fundamental of $SU(2)$

$$Y = -\frac{1}{2} \mathbf{1}_{2 \times 2}$$

$$\begin{aligned} Q L_e &= \left(\frac{\sigma^3}{2} - \frac{1}{2} \mathbf{1} \right) L_e = \frac{1}{2} \left[\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right] L_e = \\ &= \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \nu_{eL} \\ e_L \end{pmatrix} \end{aligned}$$

From these eigenvalue equations we read off that

$$\begin{cases} \text{the electric charge of the electron is } -1 \\ \text{the neutrino is electrically neutral} \end{cases}$$

[p. 21*****]

2) The quark fields are charged also under $SU(3)_c$

For instance, if we consider the quark doublet

$$Q_u = \begin{pmatrix} u_L \\ d_L \end{pmatrix}$$

Both u_L and d_L are **triplet** under $SU(3)_c$

However, since we are typically interested in **weak interactions**, we omit this extra structure.

Added explanation (not in the original notes) — Checking every hypercharge in the table, and where the $\frac{1}{6}$ comes from

The claim of comment 1) is that the hypercharges were *chosen* to make $Q = T^3 + Y$ come out right. Here is the full check — it takes one line per field.

field	rep	T^3	Y	$Q = T^3 + Y$	observed
u_L	$(\mathbf{3}, \mathbf{2}, \frac{1}{6})$	$+\frac{1}{2}$	$\frac{1}{6}$	$+\frac{2}{3}$	$+\frac{2}{3} \checkmark$
d_L	$(\mathbf{3}, \mathbf{2}, \frac{1}{6})$	$-\frac{1}{2}$	$\frac{1}{6}$	$-\frac{1}{3}$	$-\frac{1}{3} \checkmark$
u_R	$(\mathbf{3}, \mathbf{1}, \frac{2}{3})$	0	$\frac{2}{3}$	$+\frac{2}{3}$	$+\frac{2}{3} \checkmark$
d_R	$(\mathbf{3}, \mathbf{1}, -\frac{1}{3})$	0	$-\frac{1}{3}$	$-\frac{1}{3}$	$-\frac{1}{3} \checkmark$
ν_{eL}	$(\mathbf{1}, \mathbf{2}, -\frac{1}{2})$	$+\frac{1}{2}$	$-\frac{1}{2}$	0	0 $\checkmark$
e_L	$(\mathbf{1}, \mathbf{2}, -\frac{1}{2})$	$-\frac{1}{2}$	$-\frac{1}{2}$	-1	-1 $\checkmark$
e_R	$(\mathbf{1}, \mathbf{1}, -1)$	0	-1	-1	-1 $\checkmark$

The non-trivial consistency. e_L and e_R have *different* T^3 and *different* Y , yet the *same* Q . That is not automatic — it is forced by $Y(e_R) = Y(e_L) + T^3(e_L)$. If it failed, the photon would couple differently to the two chiralities and QED would violate parity. *The hypercharge assignments are exactly what*

makes electromagnetism come out vector-like from a chiral theory, which is the consistency check anticipated at the end of Lecture 31.

Where $\frac{1}{6}$ comes from. It is the average of the two quark charges: $Y(Q_u) = \frac{1}{2}(\frac{2}{3} - \frac{1}{3}) = \frac{1}{6}$, since $T^3 = \pm\frac{1}{2}$ averages to zero over a doublet. The strange-looking value is a consequence of quarks having fractional charges, not an independent oddity.

Why colour can be dropped. $SU(3)_c$ commutes with everything electroweak, and the Higgs is a colour singlet, so colour is a spectator in every formula from [p. 21*****] to [p. 34*****]. Its only role is as a multiplicity factor of 3 — which matters in exactly two places: anomaly cancellation, and the W and Z decay widths.

PART II — The electroweak currents [p. 21*****]–[p. 27*****]

4 The kinetic Lagrangians

[p. 21*****]–[p. 22*****]

■ ELECTROWEAK CURRENTS

Once we know the transformation properties of the fermions, we can write the **kinetic Lagrangian LEPTONS**

$$\begin{aligned}\mathcal{L}_{\text{kin}} = & i \bar{L}_e \gamma^\mu \left(\partial_\mu + i g_L W_\mu^A \frac{\sigma^A}{2} - i \frac{g_Y}{2} B_\mu \right) L_e \\ & + i \bar{e}_R \gamma^\mu \left(\partial_\mu - i g_Y B_\mu \right) e_R\end{aligned}$$

QUARKS

$$\begin{aligned}\mathcal{L}_{\text{kin}} = & i \bar{Q}_u \gamma^\mu \left(\partial_\mu + i g_L W_\mu^A \frac{\sigma^A}{2} + i \frac{1}{6} g_Y B_\mu \right) Q_u \\ & + i \bar{u}_R \gamma^\mu \left(\partial_\mu + i \frac{2}{3} g_Y B_\mu \right) u_R + i \bar{d}_R \gamma^\mu \left(\partial_\mu - i \frac{1}{3} g_Y B_\mu \right) d_R\end{aligned}$$

[p. 22*****]

↔ N.B. we **do not include interactions with gluons**

Let us extract the interactions. We work out explicitly the case of leptons

$$\begin{aligned}\mathcal{L}_{\text{int}} = & - \bar{L}_e \gamma^\mu \left(g_L W_\mu^A \frac{\sigma^A}{2} - \frac{g_Y}{2} B_\mu \right) L_e \\ & + \bar{e}_R \gamma^\mu g_Y B_\mu e_R\end{aligned}$$

■ We write

$$\begin{aligned}g_L W_\mu^A \sigma^A - g_Y B_\mu &= g_L \begin{pmatrix} W_\mu^3 & W_\mu^1 - i W_\mu^2 \\ W_\mu^1 + i W_\mu^2 & -W_\mu^3 \end{pmatrix} - g_Y \begin{pmatrix} B_\mu & 0 \\ 0 & B_\mu \end{pmatrix} = \\ &= \begin{pmatrix} g_L W_\mu^3 - g_Y B_\mu & g_L \sqrt{2} W_\mu^+ \\ g_L \sqrt{2} W_\mu^- & -g_L W_\mu^3 - g_Y B_\mu \end{pmatrix} \quad (\otimes)\end{aligned}$$

Added explanation (not in the original notes) — Two bookkeeping points before the algebra takes over

Where each hypercharge coefficient comes from. The covariant derivative is $D_\mu = \partial_\mu + i g_L W_\mu^A T^A + i g_Y B_\mu Y$, with Y the hypercharge of that field. Reading off the table of

[p. 19*****]–[p. 20*****]:

$$Y(L_e) = -\frac{1}{2} \Rightarrow -i\frac{g_Y}{2}B_\mu, \quad Y(e_R) = -1 \Rightarrow -ig_Y B_\mu,$$

$$Y(Q_u) = \frac{1}{6}, \quad Y(u_R) = \frac{2}{3}, \quad Y(d_R) = -\frac{1}{3}.$$

Every coefficient in the two boxed Lagrangians is just the hypercharge. Note also that $T^A = 0$ on the right-handed singlets — which is why W_μ^A never appears in the e_R, u_R, d_R terms. *Right-handed fermions do not feel the W .*

The missing $\frac{1}{2}$. $\mathcal{L}_{\text{int}}$ contains $g_L W_\mu^A \frac{\sigma^A}{2} - \frac{g_Y}{2} B_\mu = \frac{1}{2} [g_L W_\mu^A \sigma^A - g_Y B_\mu]$. The “we write” step expands the bracket *without* the $\frac{1}{2}$; the factor reappears on [p. 24*****], where the matrix is multiplied by $g_L/2$. Nothing is lost — just keep track.

The off-diagonal entries. With $W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \mp iW_\mu^2)$:

$$W_\mu^1 - iW_\mu^2 = \sqrt{2} W_\mu^+, \quad W_\mu^1 + iW_\mu^2 = \sqrt{2} W_\mu^-.$$

So the Pauli matrix structure $\sigma^A W^A$ automatically produces the charge-raising and charge-lowering fields in the off-diagonal slots. *The doublet structure and the charged currents are the same fact.*

5 Rewriting in the mass eigenstates

[p. 23*****]–[p. 24*****]

We write
$$\begin{cases} g_L = c_W (g_L^2 + g_Y^2)^{1/2} \\ g_Y = s_W (g_L^2 + g_Y^2)^{1/2} \end{cases}$$

so that

$$g_L W_\mu^3 - g_Y B_\mu = (g_L^2 + g_Y^2)^{1/2} (c_W W_\mu^3 - s_W B_\mu) = (g_L^2 + g_Y^2)^{1/2} Z_\mu = \frac{g_L}{c_W} Z_\mu$$

$$\rightsquigarrow g_L W_\mu^3 - g_Y B_\mu = \frac{g_L}{c_W} Z_\mu$$

$$\begin{aligned} g_L W_\mu^3 + g_Y B_\mu &= g_L (c_W Z_\mu + s_W A_\mu) + g_Y (-s_W Z_\mu + c_W A_\mu) \\ &= (g_L c_W - g_Y s_W) Z_\mu + (g_L s_W + g_Y c_W) A_\mu \\ &\quad \left[g_Y = \frac{g_L s_W}{c_W} \right] \\ &= \left(g_L c_W - \frac{g_L s_W^2}{c_W} \right) Z_\mu + (g_L s_W + g_L s_W) A_\mu \\ &= \frac{g_L}{c_W} (1 - 2s_W^2) Z_\mu + 2g_L s_W A_\mu \end{aligned}$$

$$\rightsquigarrow g_L W_\mu^3 + g_Y B_\mu = \frac{g_L}{c_W} (1 - 2s_W^2) Z_\mu + 2g_L s_W A_\mu$$

(*) =

$$g_L W_\mu^A \sigma^A - g_Y B_\mu = g_L \begin{pmatrix} \frac{Z_\mu}{c_W} & \sqrt{2} W_\mu^+ \\ \sqrt{2} W_\mu^- & -\frac{(1 - 2s_W^2)}{c_W} Z_\mu - 2s_W A_\mu \end{pmatrix}$$

■ We also write

$$\begin{aligned} g_Y B_\mu &= g_Y (-s_W Z_\mu + c_W A_\mu) \quad \left[g_Y = \frac{g_L s_W}{c_W} \right] \\ &= \frac{g_L s_W}{c_W} (-s_W Z_\mu + c_W A_\mu) = -\frac{g_L s_W^2}{c_W} Z_\mu + g_L s_W A_\mu \end{aligned}$$

$$\rightsquigarrow g_Y B_\mu = -\frac{g_L s_W^2}{c_W} Z_\mu + g_L s_W A_\mu$$

[p. 23*****]

[p. 24*****]

Added explanation (not in the original notes) — The inverse rotation, and a sanity check on the matrix

The inverse rotation used throughout. From Lecture 31 [p. 11*****], $Z_\mu = c_W W_\mu^3 - s_W B_\mu$ and $A_\mu = s_W W_\mu^3 + c_W B_\mu$. Inverting the (orthogonal) rotation:

$$W_\mu^3 = c_W Z_\mu + s_W A_\mu, \quad B_\mu = -s_W Z_\mu + c_W A_\mu.$$

These are the substitutions made on both sheets. Note that $g_L = c_W (g_L^2 + g_Y^2)^{1/2}$ and $g_Y = s_W (g_L^2 + g_Y^2)^{1/2}$ are just the identifications $c_W = g_L / (g_L^2 + g_Y^2)^{1/2}$, $s_W = g_Y / (g_L^2 + g_Y^2)^{1/2}$ read backwards.

Check of the trigonometric step. $g_L c_W - g_Y s_W = g_L c_W - g_L s_W^2 / c_W = g_L (c_W^2 - s_W^2) / c_W = g_L (1 - 2s_W^2) / c_W \checkmark$, using $c_W^2 = 1 - s_W^2$. And $g_L s_W + g_Y c_W = g_L s_W + g_L s_W = 2g_L s_W \checkmark$ — the two contributions are *equal*, which is the algebraic reason the photon coupling ends up symmetric between the two gauge factors.

Reading the matrix. With the overall g_L pulled out:

entry	meaning
$(1, 1) = Z_\mu / c_W$	the Z coupling to ν_L — <i>no photon</i> , as it must be ($Q_\nu = 0$)
$(1, 2), (2, 1) = \sqrt{2} W_\mu^\pm$	the charged current, turning $e_L \leftrightarrow \nu_L$
$(2, 2)$	the Z and γ couplings to e_L

The absence of A_μ from the $(1, 1)$ entry is a genuine prediction. It happened because $g_L W^3 - g_Y B$ is exactly proportional to Z with no A piece — which traces back to $\det M^2 = 0$, i.e. to $Q H_0 = 0$. The neutrino is neutral because the photon is massless; the same equation says both.

6 Splitting into the three currents

[p. 24*****]–[p. 27*****]

We thus rewrite the interaction Lagrangian as follows

$$\begin{aligned}
 \mathcal{L}_{\text{int}} &= -(\bar{\nu}_{eL}, \bar{e}_L) \gamma^\mu \frac{g_L}{2} \begin{pmatrix} \frac{Z_\mu}{c_W} & \sqrt{2} W_\mu^+ \\ \sqrt{2} W_\mu^- & -\frac{(1-2s_W^2)}{c_W} Z_\mu - 2s_W A_\mu \end{pmatrix} \begin{pmatrix} \nu_{eL} \\ e_L \end{pmatrix} \\
 &\quad + \bar{e}_R \gamma^\mu \left(-\frac{g_L s_W^2}{c_W} Z_\mu + g_L s_W A_\mu \right) e_R = \\
 &= -(\bar{\nu}_{eL}, \bar{e}_L) \gamma^\mu \frac{g_L}{2} \begin{pmatrix} \frac{Z_\mu}{c_W} \nu_{eL} + \sqrt{2} W_\mu^+ e_L \\ \sqrt{2} W_\mu^- \nu_{eL} - 2s_W A_\mu e_L - \frac{(1-2s_W^2)}{c_W} Z_\mu e_L \end{pmatrix} + \\
 &\quad + g_L s_W \bar{e}_R \gamma^\mu A_\mu e_R - \frac{g_L s_W^2}{c_W} \bar{e}_R \gamma^\mu Z_\mu e_R \\
 &= \underbrace{-\frac{g_L}{2c_W} \bar{\nu}_{eL} \gamma^\mu Z_\mu \nu_{eL}}_{\text{NC}} \underbrace{-\frac{g_L}{\sqrt{2}} \bar{\nu}_{eL} \gamma^\mu W_\mu^+ e_L - \frac{g_L}{\sqrt{2}} \bar{e}_L \gamma^\mu W_\mu^- \nu_{eL}}_{\text{CC}} \\
 &\quad + \underbrace{g_L s_W \bar{e}_L \gamma^\mu A_\mu e_L}_{\text{EM}} + \underbrace{\frac{g_L(1-2s_W^2)}{2c_W} \bar{e}_L \gamma^\mu Z_\mu e_L}_{\text{NC}} \\
 &\quad + \underbrace{g_L s_W \bar{e}_R \gamma^\mu A_\mu e_R}_{\text{EM}} - \underbrace{\frac{g_L s_W^2}{c_W} \bar{e}_R \gamma^\mu Z_\mu e_R}_{\text{NC}}
 \end{aligned}$$

[p. 25*****]

We separate three terms.

■ ELECTROMAGNETIC CURRENT

$$\begin{aligned}
 \mathcal{L}_{\text{EM}} &= g_L s_W \bar{e}_L \gamma^\mu A_\mu e_L + g_L s_W \bar{e}_R \gamma^\mu A_\mu e_R = \\
 &= g_L s_W A_\mu \underbrace{(\bar{e}_L \gamma^\mu e_L + \bar{e}_R \gamma^\mu e_R)}_{(\dagger)} = g_L s_W A_\mu \bar{e} \gamma^\mu e
 \end{aligned}$$

$$(\dagger) \quad \bar{e} \gamma^\mu P_L e + \bar{e} \gamma^\mu P_R e = \bar{e} \gamma^\mu (P_L + P_R) e = \bar{e} \gamma^\mu e$$

If we identify $g_L \sin \theta_W$ with the **fundamental charge** “ e ”, we get the QED interaction

$$g_L \sin \theta_W \equiv e$$

$$\mathcal{L}_{\text{EM}} = e A_\mu \bar{e} \gamma^\mu e$$

Added explanation (not in the original notes) — This is the moment the whole construction pays off

Take stock of what has just happened. We started from a manifestly *chiral* theory in which left- and right-handed electrons live in different representations and are coupled to different gauge fields. Out of it has come

$$\mathcal{L}_{\text{EM}} = e A_\mu \bar{e} \gamma^\mu e,$$

which is **exactly the QED vertex** — vector-like, parity-conserving, one coupling constant for both chiralities.

Why it worked. $g_L s_W$ multiplies *both* the e_L and the e_R term, and it does so because $Q(e_L) = Q(e_R) = -1$ — which was arranged by the hypercharge assignments checked on [p. 20^{*****}]. Had those been chosen differently, the two coefficients would have differed, the photon coupling would have contained a γ^5 , and electromagnetism would violate parity. *It does not, to one part in 10^{12} .*

The identification $e = g_L s_W$. This is not a definition of convenience: it is a *relation between three measured quantities*. Numerically, $g_L \simeq 0.65$ and $s_W \simeq 0.47$ give $e \simeq 0.31$, i.e. $\alpha = e^2/4\pi \simeq 1/132$ — sitting, as it should, between the low-energy value $1/137$ and the running value $1/128$ at the Z mass. Equivalently $e = g_L s_W = g_Y c_W$, symmetric between the two gauge factors, as noted in Lecture 31: *the photon does not know which factor it came from.*

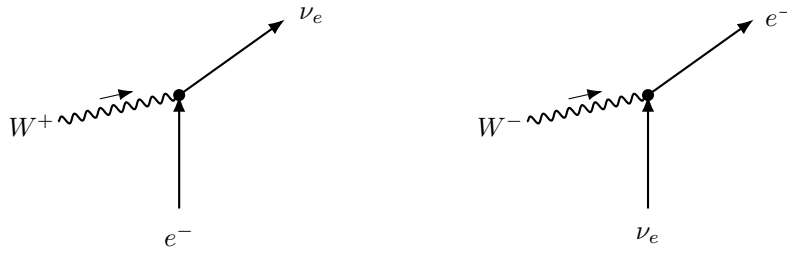
The sign. The overall minus in $\mathcal{L}_{\text{int}}$ combined with $Q_e = -1$ produces $+e A_\mu \bar{e} \gamma^\mu e$; the physical statement is that the electron has charge $-|e|$, with the sign convention absorbed into how e is defined. The notes take $e > 0$ and read the electron's charge from $Q L_e$ on [p. 20^{*****}].

■ CHARGED CURRENT

[p. 26*****]

$$\begin{aligned}\mathcal{L}_{CC} &\equiv -\frac{g_L}{\sqrt{2}} \left(\bar{\nu}_e \gamma^\mu W_\mu^+ e_L + \bar{e}_L \gamma^\mu W_\mu^- \nu_e \right) \\ &= -\frac{g_L}{\sqrt{2}} \left(\bar{\nu}_e \gamma^\mu W_\mu^+ P_L e + \bar{e} \gamma^\mu W_\mu^- P_L \nu_e \right)\end{aligned}$$

$$\mathcal{L}_{CC} = -\frac{g_L}{\sqrt{2}} W_\mu^+ \bar{\nu}_e \gamma^\mu P_L e - \frac{g_L}{\sqrt{2}} W_\mu^- \bar{e} \gamma^\mu P_L \nu_e$$



■ NEUTRAL CURRENTS

$$\begin{aligned}\mathcal{L}_{NC} &= -\frac{g_L}{2c_W} \bar{\nu}_e \gamma^\mu Z_\mu \nu_e + \frac{g_L(1-2s_W^2)}{2c_W} \bar{e}_L \gamma^\mu Z_\mu e_L - \frac{g_L s_W^2}{c_W} \bar{e}_R \gamma^\mu Z_\mu e_R = \\ &= -\frac{g_L}{2c_W} Z_\mu \bar{\nu}_e \gamma^\mu P_L \nu_e + \frac{g_L}{2c_W} Z_\mu \bar{e} \gamma^\mu (1-2s_W^2) P_L e \\ &\quad - \frac{g_L}{2c_W} Z_\mu \bar{e} \gamma^\mu 2s_W^2 P_R e = \\ &= -\frac{g_L}{2c_W} Z_\mu \left[\bar{\nu}_e \gamma^\mu P_L \nu_e - \bar{e} \gamma^\mu \left((1-2s_W^2) P_L - 2s_W^2 P_R \right) e \right]\end{aligned}$$

[p. 27*****]

$$\mathcal{L}_{NC} = -\frac{g_L}{2c_W} Z_\mu \left\{ \bar{\nu}_e \gamma^\mu P_L \nu_e - \bar{e} \gamma^\mu \left[(1-2s_W^2) P_L - 2s_W^2 P_R \right] e \right\}$$

Added explanation (not in the original notes) — Reading the three currents against each other — the structure of the weak force

current	boson	chirality structure	parity
electromagnetic	A_μ	$P_L + P_R$ — pure vector	conserved
charged	$W_\mu^\pm$	P_L only — pure $V-A$	maximally violated
neutral	Z_μ	unequal mix of P_L and P_R	partially violated

The charged current is maximally parity-violating. P_L appears alone, with no P_R term at all — because e_R is an $SU(2)_L$ singlet and simply cannot couple to $W^\pm$. Writing $P_L = \frac{1}{2}(1 - \gamma^5)$, the vertex is $\propto \gamma^\mu(1 - \gamma^5)$: this is the famous “ $V-A$ ” structure of the weak interaction, inferred from experiment in the 1950s and here *derived* from the representation assignments.

The neutral current is the odd one out, and that is why it was hard to find. Its coefficients,

$$g_L^e = -\frac{1}{2} + s_W^2, \quad g_R^e = +s_W^2,$$

(reading off $-\frac{1}{2}[(1-2s_W^2)P_L - 2s_W^2P_R]$) are *neither* equal (as for the photon) *nor* is one of them zero (as for the W). They depend on θ_W — so measuring neutral-current interactions *measures the Weinberg angle*. Historically this worked in reverse: neutral currents were predicted by this Lagrangian, discovered at CERN in 1973 (Gargamelle), and their rates fixed s_W^2 well enough to predict M_W and M_Z before the W and Z were produced in 1983.

A numerical accident worth knowing. With $s_W^2 \simeq 0.23$, the left-handed electron coupling is $-\frac{1}{2} + 0.23 \simeq -0.27$ while the right-handed one is $+0.23$: nearly equal in magnitude. And the *vector* coupling $g_V = g_L + g_R = -\frac{1}{2} + 2s_W^2 \simeq -0.04$ is *nearly zero*. The Z almost decouples from the electron's vector current — which is why parity-violating asymmetries in e^+e^- physics are small and had to be measured very precisely.

And the neutrino. $\bar{\nu}\gamma^\mu P_L\nu$ with coefficient $-\frac{1}{2}$: the neutrino couples to the Z but to nothing else. That is what makes neutrino detection so hard, and what makes the invisible Z width a measurement of the *number of light neutrino species* — famously $N_\nu = 2.984 \pm 0.008$, i.e. exactly three generations.

PART III — Applications: muon decay and beta decay [p. 27***]–[p. 29*****]**

7 The muon decay

[p. 27*****]–[p. 29*****]

APPLICATION OF CHARGED CURRENT: THE MUON DECAY

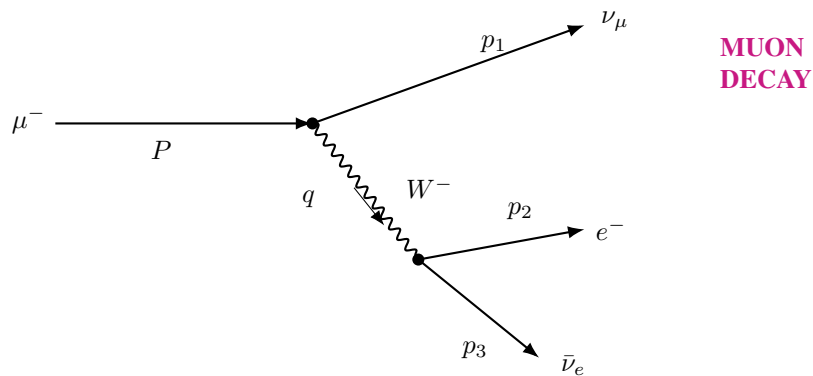
The very same computation that led to Eq. ★ can be repeated also for the other $SU(2)_L$ doublets. We can thus write

$$\begin{aligned} \mathcal{L}_{CC} = & -\frac{g_L}{\sqrt{2}} W_\mu^+ \bar{\nu}_e \gamma^\mu P_L e + \text{h.c.} + \\ & -\frac{g_L}{\sqrt{2}} W_\mu^+ \bar{\nu}_\mu \gamma^\mu P_L \mu + \text{h.c.} + \\ & -\frac{g_L}{\sqrt{2}} W_\mu^+ \bar{u} \gamma^\mu P_L d + \text{h.c.} \end{aligned} \quad \left. \begin{array}{l} \} (\otimes) \\ \} (\odot) \end{array} \right\}$$

Added explanation (not in the original notes) — the two circled symbols

The original marks the two braces with two very similar circled stars. They are distinguished here as $(\otimes)$ for the two *leptonic* lines — which is what [p. 27*****] invokes for muon decay — and $(\odot)$ for the *quark* line, which [p. 29*****] invokes for beta decay.

Interactions in $(\otimes)$ open up the possibility of having the following decay process:



Very simple kinematics implies (working in the **rest frame** of the decaying muon)

I $P^\mu = (m_\mu, \vec{0})$

II $P = p_1 + q \Rightarrow q = P - p_1 \Rightarrow q^2 = P^2 + \underbrace{p_1^2}_{=0} - 2P \cdot p_1 = m_\mu^2 - 2p_1^0 m_\mu \leq \underbrace{m_\mu^2}_{\text{since } p_1^0 > 0}$

(massless neutrino)

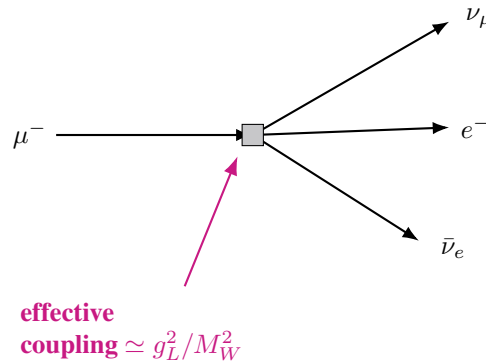
III $q = p_2 + p_3 \Rightarrow q^2 = \underbrace{p_2^2}_{=0} + \underbrace{p_3^2}_{=0} + 2p_2 \cdot p_3 = 2p_2^0 p_3^0 (1 - \cos \theta) \geq 0$

(neglecting the electron mass)

[p. 28*****]

$$0 \leq q^2 \leq m_\mu^2 \ll M_W^2$$

We thus approximate



Let us estimate the **decay width** for this process, using **dimensional analysis**. The decay width Γ has mass dimension $[\Gamma] = [M]$

Added explanation (not in the original notes) — Why the kinematic bounds matter, and what the “effective coupling” really is

The two inequalities, and why both are needed. Step II gives an *upper* bound $q^2 \leq m_\mu^2$; step III gives a *lower* bound $q^2 \geq 0$. Together they confine the momentum flowing through the W propagator to a window of width $m_\mu^2 \simeq 0.011 \text{ GeV}^2$, against $M_W^2 \simeq 6460 \text{ GeV}^2$. The ratio is $\sim 2 \times 10^{-6}$: the W is *extremely* far off shell.

Hence the contraction. The propagator

$$\frac{-i(g^{\mu\nu} - q^\mu q^\nu / M_W^2)}{q^2 - M_W^2} \xrightarrow{q^2 \ll M_W^2} \frac{i g^{\mu\nu}}{M_W^2}$$

loses all q dependence. In position space that is a **contact interaction**: the two vertices collapse to a single point, and the amplitude becomes $\sim g_L^2/M_W^2$ times a product of four fermion fields. This is exactly *Fermi's* 1934 four-fermion theory — and the identification is

$$\frac{G_F}{\sqrt{2}} = \frac{g_L^2}{8M_W^2} = \frac{1}{2v^2},$$

using $M_W = \frac{1}{2}g_L v$. Fermi's constant is nothing but $1/v^2$ in disguise.

What this teaches, in EFT language. Below M_W the correct description is not the Standard Model but an effective theory of four-fermion operators, with the heavy W integrated out. That is precisely the logic of Lectures 25–29 applied in reverse: Fermi's theory is the low-energy EFT, non-renormalisable, with cutoff $\sim M_W$, and the Standard Model is its UV completion. *Historically it went the other way* — Fermi's theory came first, its breakdown at high energy was the clue, and the W was the resolution.

Since $\Gamma \propto |\mathcal{M}|^2$, the amplitude squared, it will contain for sure the factor $\frac{g_L^4}{M_W^4}$

[p. 29*****]

Since we take $m_e = 0$, the only way to compensate the dimensionful factor $1/M_W^4$ and get something with the right dimension, is to multiply by m_μ^5 . We get

$$\Gamma_\mu \simeq \frac{g_L^4}{M_W^4} m_\mu^5 \simeq \frac{m_\mu^5}{v^4} \longrightarrow v = 246 \text{ GeV}$$

From the **precise measurement of the muon decay width**, we measured the value of “ v ”

Added explanation (not in the original notes) — The dimensional-analysis estimate, checked against the real formula

The counting. $[\Gamma] = [M]$. The squared amplitude carries g_L^4/M_W^4 , which has dimension $[M]^{-4}$. With $m_e = m_\nu = 0$ the *only* mass left in the problem is m_μ , so the missing $[M]^5$ must be m_μ^5 . No other answer is possible — this is dimensional analysis doing real work.

Why $g_L^4/M_W^4 \simeq 1/v^4$. With $M_W = \frac{1}{2}g_L v$:

$$\frac{g_L^4}{M_W^4} = \frac{g_L^4}{(\frac{1}{2}g_L v)^4} = \frac{16}{v^4}.$$

The gauge coupling *cancels completely*. Muon decay does not measure g_L ; it measures v . That is the point of the green remark, and it is why v is the best-known parameter of the electroweak sector.

Against the exact tree-level result. The full calculation gives

$$\Gamma_\mu = \frac{G_F^2 m_\mu^5}{192\pi^3}, \quad G_F = \frac{1}{\sqrt{2}v^2} \implies \Gamma_\mu = \frac{m_\mu^5}{384\pi^3 v^4} \simeq \frac{1}{1.19 \times 10^4} \cdot \frac{m_\mu^5}{v^4}.$$

So the estimate has the right *parametric* form and misses only the numerical factor $1/(384\pi^3)$ — the usual phase-space and π 's. Numerically $\tau_\mu = 1/\Gamma_\mu = 2.19 \times 10^{-6} \text{ s}$ against the measured $2.197 \times 10^{-6} \text{ s}$ — agreement to half a per cent. Inverting: $v = 246.2 \text{ GeV}$.

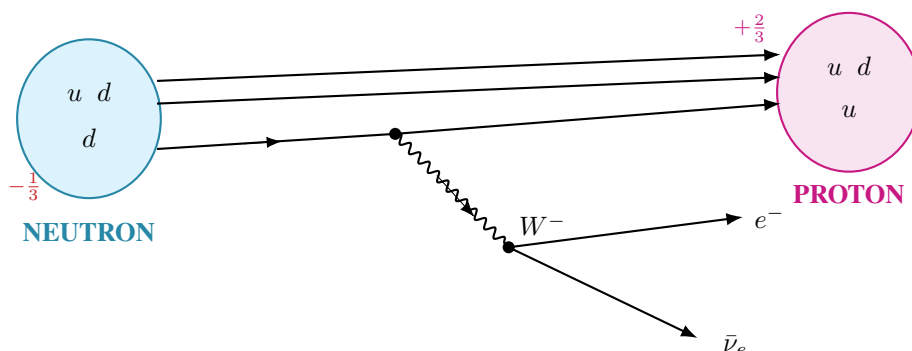
The m_μ^5 is not an accident of this process. Any weak decay with a light final state scales as (Fermi constant) $^2 \times$ (available energy) 5 ; this is why the neutron, with only $\sim 1.3 \text{ MeV}$ of energy release, lives for $\sim 880 \text{ s}$ while the muon with 106 MeV lives for microseconds. Fifth powers are steep.

8 Neutron beta decay

[p. 29*****]

APPLICATION OF CHARGED CURRENT: THE BETA DECAY

The interaction in (⊙) explains the **beta decay of the neutron**, $n \rightarrow p + e^- + \bar{\nu}_e$



Added explanation (not in the original notes) — What beta decay is, at the quark level — and why this was the original puzzle

The process, unpacked. A neutron is udd and a proton is uud . Beta decay is therefore, at the fundamental level, a single quark transition

$$d \longrightarrow u + W^-, \quad W^- \rightarrow e^- + \bar{\nu}_e,$$

with the other two quarks along for the ride (“spectators”). *The same vertex $\bar{u}\gamma^\mu P_L d$ that appears in (⊙) does the whole job.* Notice that the electric charge bookkeeping works: $-\frac{1}{3} \rightarrow +\frac{2}{3} + (-1) \checkmark$.

Why this is the same physics as muon decay. Both are the W exchanged far off shell between two charged currents, so both are governed by the same $G_F = 1/(\sqrt{2}v^2)$. The universality of the weak coupling — that the muon and the neutron decay with the *same* strength — is a prediction, and its confirmation was one of the first successes of the theory. (The small deviation, $\sim 5\%$, is the CKM element $|V_{ud}| = 0.974$: real quarks mix across generations, which the one-generation treatment here suppresses.)

Historically, this process started everything. The continuous electron spectrum in beta decay led Pauli to postulate the neutrino in 1930; Fermi’s 1934 four-fermion theory was written to describe exactly this diagram in its contracted form; the observation that it violates parity (Wu, 1957) forced the $V-A$ structure; and the non-renormalisability of the contact theory demanded the W . *Everything in Lectures 24–32 can be read as the answer to the question “why does the neutron decay?”*

And one cosmological consequence. The neutron lifetime, $\tau_n \simeq 879$ s, sets how many neutrons survive to be captured into helium in the first few minutes of the universe — so this diagram fixes the primordial helium abundance at about 25% by mass. Change v and you change the chemical composition of the universe.

PART IV — The generation of mass [p. 30*****]–[p. 34*****]

9 Why a chiral theory forbids fermion masses

[p. 30*****]

■ THE GENERATION OF MASS

The theory has a **problem**.We cannot write a mass term for the fermion fields since the theory is chiral

[p. 30*****]

In fact, let us rewrite a typical mass term $\bar{\Psi}\Psi$ for a Dirac field as follows.

$$\Psi = \Psi_L + \Psi_R ; \quad \bar{\Psi} = \bar{\Psi}_L + \bar{\Psi}_R$$

$$\Rightarrow \quad \bar{\Psi}\Psi = (\bar{\Psi}_L + \bar{\Psi}_R)(\Psi_L + \Psi_R) = \bar{\Psi}_L\Psi_R + \bar{\Psi}_R\Psi_L$$

Terms of the kind $\bar{\Psi}_L\Psi_L$ and $\bar{\Psi}_R\Psi_R$ **identically vanish**.

This is evident if we write, for instance

$$\Psi_L = \begin{pmatrix} \psi_L \\ 0 \end{pmatrix} \Rightarrow \Psi_L^\dagger = (\psi_L^\dagger, 0) \Rightarrow \bar{\Psi}_L = \Psi_L^\dagger \gamma^0 = (\psi_L^\dagger, 0) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = (0, \psi_L^\dagger)$$

and we thus have $\bar{\Psi}_L\Psi_L = 0$ The problem arises because if ψ_L and ψ_R transform **differently**, ★ is not invariant under the gauge transformations.The presence of the **Higgs** solves the problem.

Consider, for simplicity, the case of one generation. Let us focus on the leptonic sector.

Added explanation (not in the original notes) — The two separate obstructions — keep them apart

This sheet packs two logically distinct facts into one page, and it is easy to conflate them.

Obstruction 1 (kinematic, and unavoidable): $\bar{\Psi}_L\Psi_L = 0$. This is pure Dirac algebra and holds in *any* theory, gauge or not. In the chiral (Weyl) basis $\gamma^0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, so $\bar{\Psi}_L$ has support only in the lower two components while Ψ_L has support only in the upper two. Their product vanishes identically. Equivalently, using the projectors:

$$\bar{\Psi}_L\Psi_L = \Psi^\dagger P_L^\dagger \gamma^0 P_L \Psi = \bar{\Psi} P_R P_L \Psi = 0,$$

since $\gamma^0 P_L = P_R \gamma^0$ and $P_R P_L = 0$. A mass term must always pair opposite chiralities. There is no way around this.**Obstruction 2 (dynamical, and the real problem): gauge invariance.** Having established that $\bar{\Psi}\Psi = \bar{\Psi}_L\Psi_R + \text{h.c.}$, one must ask whether *that* is allowed. In the Standard Model it is not. Take the electron: $\bar{e}_L$ transforms as **2** of $SU(2)_L$ with $Y = +\frac{1}{2}$, while e_R is a **1** with $Y = -1$. The product is a **2** with $Y = -\frac{1}{2}$: *not a singlet*, therefore not gauge invariant, therefore forbidden in the Lagrangian.**The scale of the problem.** This is not a technicality about one field. *Every* fermion in the Standard Model is forbidden a mass by the same argument — quarks and leptons, all three generations. Written naively, the theory predicts twelve massless fermions plus a massless photon and Z and W . Reality contains one massless particle (the photon). The Higgs is what closes the entire gap.

The resolution in one line. $\bar{L}_e H e_R$ carries $Y = +\frac{1}{2} + \frac{1}{2} - 1 = 0$ and is an $SU(2)$ singlet ($\mathbf{2} \otimes \mathbf{2} \supset \mathbf{1}$). The Higgs doublet has exactly the quantum numbers of the mismatch. That is not a coincidence engineered afterwards: $H \sim (\mathbf{1}, \mathbf{2}, \frac{1}{2})$ was fixed on Lecture 30 [p. 2*****] by requiring a neutral vev, and it happens to be precisely what the Yukawa couplings need.

10 The leptonic Yukawa coupling

[p. 31*****]

$$\begin{aligned} H &\sim (1, 2, \tfrac{1}{2}) \\ L_e &\sim (1, 2, -\tfrac{1}{2}) \\ e_R &\sim (1, 1, -1) \end{aligned}$$

} Using these fields it is possible to write the **invariant operators**

$$\mathcal{L}_Y = -y_e \bar{e}_R H^\dagger L_e + \text{h.c.}$$

[p. 31*****]

After SSB, we write

$$\begin{aligned} \mathcal{L}_Y &= -y_e \bar{e}_R \frac{1}{\sqrt{2}} (0, v+h) \begin{pmatrix} \nu_{eL} \\ e_L \end{pmatrix} + \text{h.c.} = \\ &= -y_e \bar{e}_R \frac{1}{\sqrt{2}} (v+h) e_L + \text{h.c.} = -\frac{y_e v}{\sqrt{2}} \bar{e}_R e_L - \frac{y_e}{\sqrt{2}} h \bar{e}_R e_L + \text{h.c.} \\ &= \underbrace{-\frac{y_e v}{\sqrt{2}} (\bar{e}_R e_L + \bar{e}_L e_R)}_{(1)} - \underbrace{\frac{y_e}{\sqrt{2}} h (\bar{e}_R e_L + \bar{e}_L e_R)}_{(2)} \end{aligned}$$

(1) **ordinary mass term** for the electron, with mass

(2) **coupling between the Higgs and the electron field,**

$$m_e = \frac{y_e v}{\sqrt{2}}$$

$$\mathcal{L}_{\text{int}} = -\frac{m_e}{v} h \bar{e} e$$

OBS: the neutrino remains massless

“ y_e ” is called electron Yukawa coupling. In the SM, it is a **free parameter** that we fix with experiments. Since $m_e \simeq 0.5 \text{ MeV}$, from $v = 246 \text{ GeV}$ we get

$$y_e \simeq 10^{-6}$$

Added explanation (not in the original notes) — Two predictions and one non-prediction

The non-prediction. y_e is a free parameter. The Standard Model does not explain why the electron weighs 0.511 MeV ; it only *allows* it to. Precisely: $y_e = \sqrt{2} m_e / v = \sqrt{2} \times 0.511 \times 10^{-3} / 246.2 = 2.9 \times 10^{-6}$. The notes quote $\simeq 10^{-6}$; same order.

Prediction 1: the Higgs–fermion coupling is fixed by the mass. Terms (1) and (2) come from the *same* y_e , so the ratio is not free:

$$\mathcal{L}_{\text{int}} = -\frac{m_e}{v} h \bar{e} e.$$

Every fermion couples to the Higgs in proportion to its own mass, with the universal constant $1/v$. This is the sharpest statement the theory makes about the Higgs, it is parameter-free once the masses are known, and it is exactly what the plot on [p. 35*****] tests. It also explains the LHC’s search strategy: $h \rightarrow b\bar{b}$ and $h \rightarrow \tau\tau$ are seen, $h \rightarrow \mu\mu$ is at the edge, $h \rightarrow ee$ is hopeless — the rates go as m_f^2 .

Prediction 2: the neutrino is exactly massless. Not approximately — *exactly*, because there is no ν_R to pair ν_L with, and no operator of dimension four can be written. **This prediction is false.** Neutrino oscillations (Super-Kamiokande 1998, SNO 2002) require at least two non-zero masses, of order $0.01\text{--}0.1 \text{ eV}$. This is the one place where the Standard Model as constructed in these lectures is known to be incomplete.

How it is usually fixed. Either add $\nu_R \sim (\mathbf{1}, \mathbf{1}, 0)$ and a Yukawa $y_\nu \sim 10^{-12}$ (which works but is ugly), or write the dimension-five Weinberg operator $\frac{c}{\Lambda} (\bar{L}^c \tilde{H})(\tilde{H}^\dagger L)$, which gives $m_\nu \sim v^2/\Lambda$ and, for $m_\nu \sim 0.05$ eV, points to $\Lambda \sim 10^{14}$ GeV. The second is more interesting: *neutrino masses are the one experimental handle on physics far above the electroweak scale.*

11 The quark sector: down and up

[p. 32*****]–[p. 34*****]

Let us quickly discuss the **quark sector**. We have

[p. 32*****]

$$\begin{cases} Q_u \sim (\mathbf{3}, \mathbf{2}, \frac{1}{6}) \\ u_R \sim (\mathbf{3}, \mathbf{1}, \frac{2}{3}) \\ d_R \sim (\mathbf{3}, \mathbf{1}, -\frac{1}{3}) \\ H \sim (\mathbf{1}, \mathbf{2}, \frac{1}{2}) \end{cases}$$

Following what we have done in the leptonic sector, we construct the operator

$$\mathcal{L}_Y^d = -y_d \bar{d}_R H^\dagger Q_u + \text{h.c.}$$

invariant under $SU(3)_c \times SU(2)_L \times U(1)_Y$

In the broken phase

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}; \quad \bar{d}_R H^\dagger Q_u = \bar{d}_R \frac{1}{\sqrt{2}} (0, h + v) \begin{pmatrix} u_L \\ d_L \end{pmatrix} = \frac{1}{\sqrt{2}} (h + v) \bar{d}_R d_L$$

$$\mathcal{L}_Y = \underbrace{-\frac{y_d v}{\sqrt{2}} \bar{d} d}_{(1)} - \underbrace{\frac{y_d}{\sqrt{2}} h \bar{d} d}_{(2)}$$

(1) **ordinary mass term** for the down quark

(2) **interaction with the Higgs**

$$m_d = \frac{y_d v}{\sqrt{2}}$$

$$\mathcal{L}_{\text{int}} = -\frac{m_d}{v} h \bar{d} d$$

$$y_d \simeq 2 \times 10^{-5}$$

There is one last problem. The up quark remains massless

↔ To fix this problem, we use a property of $SU(2)$: if H transforms as a doublet, also $\tilde{H} = i\sigma_2 H^*$ transforms as a doublet

[p. 33*****]

Let us write $\varepsilon \equiv i\sigma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and $\tilde{H} = \varepsilon H^*$

If $H \rightarrow UH$, then $H^* \rightarrow U^* H^*$, and consequently $\tilde{H} = \varepsilon H^* \rightarrow \varepsilon U^* H^*$

In $SU(2)$, we have

$$\varepsilon U^* = U \varepsilon$$

so that $\tilde{H} \rightarrow \varepsilon U^* H^* = U \varepsilon H^* = U \tilde{H}$

We thus have

$$\tilde{H} = i\sigma_2 H^* \sim (\mathbf{1}, \mathbf{2}, -\frac{1}{2})$$

and we can construct

$$\mathcal{L}_Y^u = -y_u \bar{u}_R \tilde{H}^\dagger Q_u + \text{h.c.}$$

invariant under $SU(3)_c \times SU(2)_L \times U(1)_Y$

In the broken phase

$$\tilde{H} = i\sigma_2 \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v+h \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v+h \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} v+h \\ 0 \end{pmatrix}$$

$$\mathcal{L}_Y^u = -y_u \bar{u}_R \frac{1}{\sqrt{2}} (v+h, 0) \begin{pmatrix} u_L \\ d_L \end{pmatrix} + \text{h.c.} =$$

$$\mathcal{L}_Y^u = \underbrace{-\frac{y_u v}{\sqrt{2}} \bar{u}u}_{\text{mass term}} - \underbrace{\frac{y_u}{\sqrt{2}} h \bar{u}u}_{\text{interaction}}$$

[p. 34*****]

$$m_u = \frac{y_u v}{\sqrt{2}}$$

$$\mathcal{L}_{\text{int}} = -\frac{m_u}{v} h \bar{u}u$$

Added explanation (not in the original notes) — Why $\tilde{H}$ works — the pseudo-reality of $SU(2)$

The problem, precisely. $\bar{u}_R H^\dagger Q_u$ would carry hypercharge $-\frac{2}{3} - \frac{1}{2} + \frac{1}{6} = -1 \neq 0$: forbidden. What is needed is a doublet with $Y = -\frac{1}{2}$. In a generic gauge group one would have to add a *second* scalar doublet with the opposite hypercharge. $SU(2)$ lets us avoid that.

The identity $\varepsilon U^* = U \varepsilon$. For $U = e^{-i\alpha_A \sigma^A/2} \in SU(2)$ this follows from $\sigma_2 \sigma^A \sigma_2 = -\sigma^A$ (check: $\sigma^{1,3}$ are real and anticommute with σ_2 ; σ^2 is imaginary and commutes with itself). Hence $\varepsilon U^* \varepsilon^{-1} = U$, i.e. $\mathbf{2}^* \cong \mathbf{2}$: the doublet is **pseudo-real**. Complex conjugation gives back the same representation, up to the similarity transformation ε .

The hypercharge flips because of the conjugation. H^* has $Y = -\frac{1}{2}$, and ε is a fixed numerical matrix carrying no charge. So $\tilde{H} \sim (\mathbf{1}, \mathbf{2}, -\frac{1}{2})$ and $\bar{u}_R \tilde{H}^\dagger Q_u$ has $-\frac{2}{3} + \frac{1}{2} + \frac{1}{6} = 0 \checkmark$.

The neat consequence. $\tilde{H}$ picks out the *upper* component of the doublet, $\tilde{H} \rightarrow \frac{1}{\sqrt{2}}(v+h, 0)^T$, so it pairs u_R with u_L rather than d_L . One Higgs doublet gives masses to *both* the up-type and down-type fermions. **This is why the Standard Model needs exactly one Higgs doublet, no more.**

And why some theories give it up. In supersymmetry the superpotential must be holomorphic — H^* is not allowed — so two doublets H_u and H_d are required, and the Higgs sector has five physical scalars instead of one. The trick used here is special to non-supersymmetric $SU(2)$.

Numbers. $y_d = \sqrt{2} m_d/v$ with $m_d \simeq 4.7$ MeV gives 2.7×10^{-5} (notes: $\simeq 2 \times 10^{-5} \checkmark$). For the top, $y_t = \sqrt{2} \times 172.5/246.2 = 0.99$: *the top Yukawa is essentially exactly one*. Whether that is a coincidence is one of the open questions of the subject.

PART V — Observational consequences, and the electroweak phase transition [p. 34*****]–[p. 36*****]

12 Observational consequences

[p. 34*****]–[p. 35*****]

OBSERVATIONAL CONSEQUENCES



$$v = 246 \text{ GeV}$$

← from **muon decay lifetime**



$$\begin{aligned} M_W &\simeq 80 \text{ GeV} \\ M_Z &\simeq 91 \text{ GeV} \end{aligned}$$



$$\begin{aligned} g_L &\simeq 0.65 \\ g_Y &\simeq 0.35 \\ \sin^2 \theta_W &\simeq 0.223 \end{aligned}$$

↑ UA1 and UA2 experiments at CERN



$$m_h \simeq 125 \text{ GeV}$$



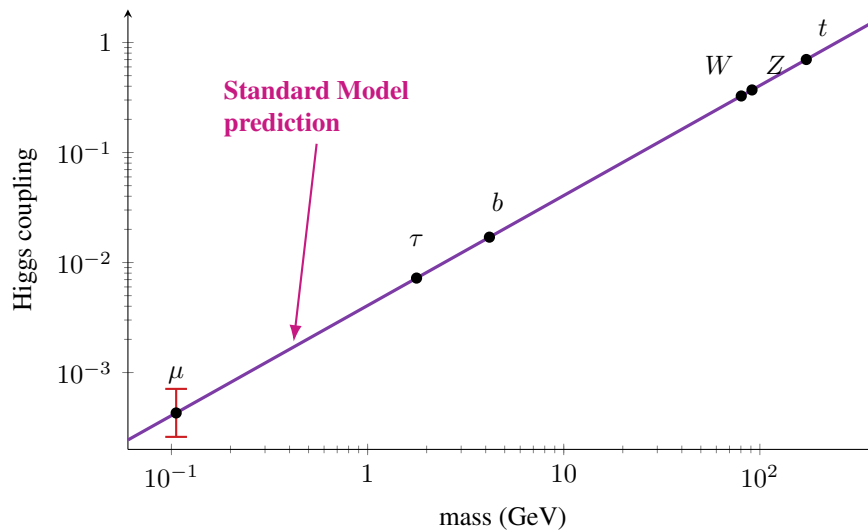
$$\lambda \simeq 0.13$$

↑ **prediction** for the strength of the Higgs self-couplings!

↑ Higgs boson discovery, 2012

Furthermore we measured (some of) the **Higgs couplings with gauge bosons and fermions**. As we have seen, in the Standard Model the electroweak symmetry breaking predicts that the Higgs coupling with a strength that is **proportional to the particle mass**

[p. 35*****]



Added explanation (not in the original notes) — reading the plot

The handwritten labels are hard to make out in the scan. The six couplings measured at the LHC are those of μ , τ , b , W , Z and t ; the figure is redrawn with those. The straight line is coupling = m/v with $v = 246.2 \text{ GeV}$ — a slope-one line in log-log, with *no free parameters*. The error bar on the muon point is the largest because $h \rightarrow \mu\mu$ is the rarest of the six.

Added explanation (not in the original notes) — What each measured number does — and which is a genuine prediction

input	measured from	year	fixes
$v = 246.2 \text{ GeV}$	muon lifetime	1940s–	the overall EW scale
M_W, M_Z	$S\bar{p}pS$ collider, CERN	1983	g_L, g_Y, θ_W
$m_h = 125.25 \text{ GeV}$	ATLAS & CMS, LHC	2012	λ

The counting. The bosonic sector has four parameters: g_L, g_Y, μ^2, λ . Three measurements (v ; M_W and M_Z ; m_h) fix all four — and there is one relation left over, $\rho = M_W^2/(M_Z^2 c_W^2) = 1$, which is therefore a *test*. It passes.

Checking the numbers. $g_L = 2M_W/v = 2(80.38)/246.2 = 0.653 \checkmark$; $\sin^2 \theta_W = 1 - M_W^2/M_Z^2 = 1 - (80.38/91.19)^2 = 0.223 \checkmark$ (the notes' value; the $\overline{\text{MS}}$ value 0.231 differs by radiative corrections); $g_Y = g_L \tan \theta_W = 0.653 \times 0.536 = 0.350 \checkmark$; $\lambda = m_h^2/(2v^2) = 125.25^2/(2 \times 246.2^2) = 0.129 \checkmark$.

Why λ is called a prediction. Before 2012, m_h was unknown and λ was a free parameter. Measuring m_h fixed $\lambda \simeq 0.13$ — and since λ also controls the *Higgs self-interactions* $-\lambda v h^3 - \frac{\lambda}{4} h^4$ (Lecture 31 [p. 15*****]), the trilinear coupling $\lambda_{hhh} = 3m_h^2/v$ is now completely predicted. It has not yet been measured: that requires double-Higgs production, and it is a headline goal of the high-luminosity LHC. *It is the last untested piece of the mechanism.*

Why the coupling plot is the decisive test. Every other measurement tests that the particles exist with the right masses. This one tests the *mechanism*: that a single field with a single vev is responsible for all of them, which is what forces coupling $\propto$ mass with the same v across five orders of magnitude. Any second Higgs doublet, any composite structure, any partial contribution from another source, and the points would scatter off the line.

13 The dynamics of electroweak symmetry breaking

[p. 35*****]–[p. 36*****]

THE DYNAMICS OF THE EW SYMMETRY BREAKING

The Higgs potential that we wrote is

$$V(H^\dagger H) = -\mu^2 H^\dagger H + \lambda (H^\dagger H)^2; \quad \mu^2, \lambda > 0$$

► This is valid at zero temperature

In a **cosmological setting**, the universe underwent a period during which it was dominated by a **thermal bath** at temperature “ T ”.

Particles in this bath interacted with the Higgs field, generating finite-temperature corrections to the Higgs potential

The **finite-temperature Higgs potential** becomes roughly

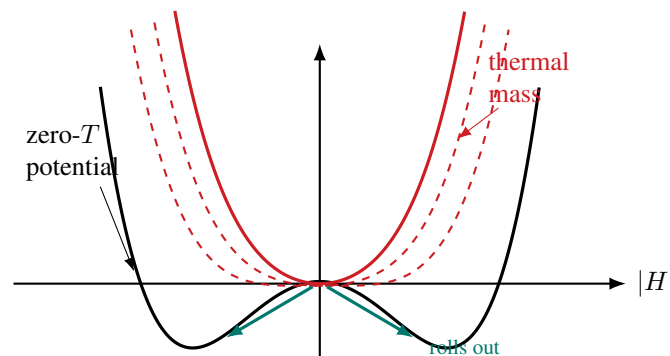
$$V(H, T) = \left(-\mu^2 + \underbrace{cT^2}_{\text{“thermal mass”}} \right) H^\dagger H + \lambda (H^\dagger H)^2$$

[p. 36*****]

Added explanation (not in the original notes) — slip flagged

The original writes the quartic as $\lambda(H^\dagger H)$, without the exponent. It must be $\lambda(H^\dagger H)^2$, as everywhere else in the notes. Corrected above.

At **high temperature**, the thermal mass **dominates** and the symmetry is **not broken**



As the universe expands and **cools down**, the Higgs starts to “feel” the Mexican hat potential, until it eventually **rolls** in the EW symmetry breaking vacuum.

This is the so-called “**ELECTROWEAK PHASE TRANSITION**”.

It happened when the universe cooled to roughly

$$T_{\text{EW}} \simeq 160 \text{ GeV} \quad (t \simeq 10^{-10} \text{ s after the Big Bang})$$

[end of the course]

Added explanation (not in the original notes) — Where the thermal mass comes from, and what the transition looks like

The physical origin of cT^2 . At temperature T the universe is filled with a plasma of W 's, Z 's, top quarks and Higgses. Every particle that couples to H contributes a loop with a thermal propagator, and the leading effect is a positive correction to the effective mass-squared:

$$c \simeq \frac{1}{16} (3g_L^2 + g_Y^2 + 4y_t^2 + 8\lambda) \simeq 0.4.$$

The dominant contributions are the top ($y_t \simeq 1$) and the gauge bosons — exactly the particles that couple most strongly to the Higgs, as the plot on [p. 35^{*****}] shows. *Symmetry restoration at high T is a completely general phenomenon* (the same reason a ferromagnet demagnetises when heated) and was first worked out by Kirzhnits and Linde, and by Dolan and Jackiw, in 1972–74.

The critical temperature. The quadratic coefficient changes sign when $cT^2 = \mu^2 = \lambda v^2$, giving

$$T_c \simeq v \sqrt{\lambda/c} \simeq 246 \times \sqrt{0.13/0.4} \simeq 140\text{--}160 \text{ GeV},$$

which is the number quoted. Converting to time via the radiation-dominated relation $t \simeq (2.4\sqrt{g_*})^{-1} M_{\text{Pl}}/T^2$ with $g_* \simeq 107$ gives $t \simeq 1.6 \times 10^{-11} \text{ s}$ — the notes' 10^{-10} s , to an order of magnitude.

But what *kind* of transition? The sketch shows a barrier developing between the symmetric and broken minima, which would make it *first order* — proceeding by bubble nucleation, out of equilibrium, with latent heat and gravitational waves. That picture requires m_h to be light. With the measured $m_h = 125 \text{ GeV}$, lattice calculations show there is **no barrier at all**: the transition is a smooth *crossover*. The two phases are continuously connected, nothing dramatic happens, and the universe simply slides into the broken phase.

Why that matters enormously. A first-order transition would provide the out-of-equilibrium condition needed for **electroweak baryogenesis** — generating the matter–antimatter asymmetry at this very moment in cosmic history. A crossover does not. So the measured Higgs mass *rules out* baryogenesis within the minimal Standard Model, and the origin of the matter in the universe remains unexplained. *The last sheet of the course ends on an open problem, and a deep one.*

Summary of Lecture 32 — the logic in twelve lines

Added explanation (not in the original notes) — Summary

Part I — the fermion content

1. Three generations, identical in every quantum number and differing only in mass. Working with one: $Q_u = (u_L, d_L)^T \sim (\mathbf{3}, \mathbf{2}, \frac{1}{6})$, $L_e = (\nu_{eL}, e_L)^T \sim (\mathbf{1}, \mathbf{2}, -\frac{1}{2})$, and the right-handed singlets u_R, d_R, e_R . **No ν_R in the original model.** [p. 19*****]–[p. 20*****]
2. The hypercharges are chosen so that $Q = T^3 + Y$ reproduces the observed electric charges — and, crucially, gives e_L and e_R the *same* Q despite different T^3 and Y . [p. 20*****]–[p. 21*****]

Part II — the currents

3. Writing $g_L W_\mu^A \sigma^A - g_Y B_\mu$ as a 2×2 matrix puts $\sqrt{2} W_\mu^\pm$ off-diagonal automatically, and rewriting the diagonal in mass eigenstates gives $g_L W^3 - g_Y B = \frac{g_L}{c_W} Z$ and $g_L W^3 + g_Y B = \frac{g_L}{c_W} (1 - 2s_W^2) Z + 2g_L s_W A$. [p. 22*****]–[p. 24*****]
4. The interaction Lagrangian then splits into three pieces. **Electromagnetic:** $\mathcal{L}_{\text{EM}} = e A_\mu \bar{e} \gamma^\mu e$ with $e \equiv g_L \sin \theta_W$ — vector-like, because $Q(e_L) = Q(e_R)$. *QED emerges unmodified from a chiral theory.* [p. 25*****]
5. **Charged:** $-\frac{g_L}{\sqrt{2}} W_\mu^+ \bar{\nu} \gamma^\mu P_L e + \text{h.c.}$ — pure P_L , i.e. $V - A$, maximal parity violation. **Neutral:** an unequal mix of P_L and P_R whose coefficients depend on θ_W . [p. 26*****]–[p. 27*****]

Part III — applications

6. Muon decay: kinematics confines $0 \leq q^2 \leq m_\mu^2 \ll M_W^2$, so the W propagator contracts to a four-fermion contact interaction of strength g_L^2/M_W^2 — Fermi's theory, recovered as the low-energy EFT. [p. 27*****]–[p. 28*****]
7. Dimensional analysis then forces $\Gamma_\mu \simeq g_L^4 m_\mu^5 / M_W^4 \simeq m_\mu^5 / v^4$: the gauge coupling cancels, so **the muon lifetime measures v** . This is how $v = 246$ GeV is known. [p. 29*****]
8. The same charged current, applied to the quark line $\bar{u} \gamma^\mu P_L d$, is neutron beta decay $n \rightarrow p e^- \bar{\nu}_e$ — one quark transition with two spectators. [p. 29*****]

Part IV — mass

9. $\bar{\Psi} \Psi = \bar{\Psi}_L \Psi_R + \text{h.c.}$, since same-chirality bilinears vanish identically. But $\bar{\Psi}_L \Psi_R$ pairs different representations, so it is *not gauge invariant*: **every Standard Model fermion is forbidden a mass.** [p. 30*****]
10. The Higgs supplies exactly the missing quantum numbers. $-y_e \bar{e}_R H^\dagger L_e$ gives, after SSB, both a mass $m_e = y_e v / \sqrt{2}$ and a coupling $-\frac{m_e}{v} h \bar{e} e$ — from the same parameter. The neutrino, having no partner, stays exactly massless. [p. 31*****]
11. The same works for d_R ; for u_R one needs a doublet of hypercharge $-\frac{1}{2}$, and $SU(2)$ supplies it for free through the pseudo-reality of the fundamental: $\tilde{H} = i\sigma_2 H^*$. *Hence one Higgs doublet suffices for the whole spectrum.* [p. 32*****]–[p. 34*****]

Part V — observations and cosmology

12. Three measurements fix everything: v (muon lifetime), M_W, M_Z (CERN 1983) giving $g_L \simeq 0.65$, $g_Y \simeq 0.35$, $\sin^2 \theta_W \simeq 0.223$, and $m_h = 125$ GeV (LHC 2012) giving $\lambda \simeq 0.13$ — which *predicts* the Higgs self-coupling. The measured Higgs couplings lie on the line m/v across five orders of magnitude. Finally, thermal corrections $-\mu^2 \rightarrow -\mu^2 + cT^2$ restore the symmetry at high T : the

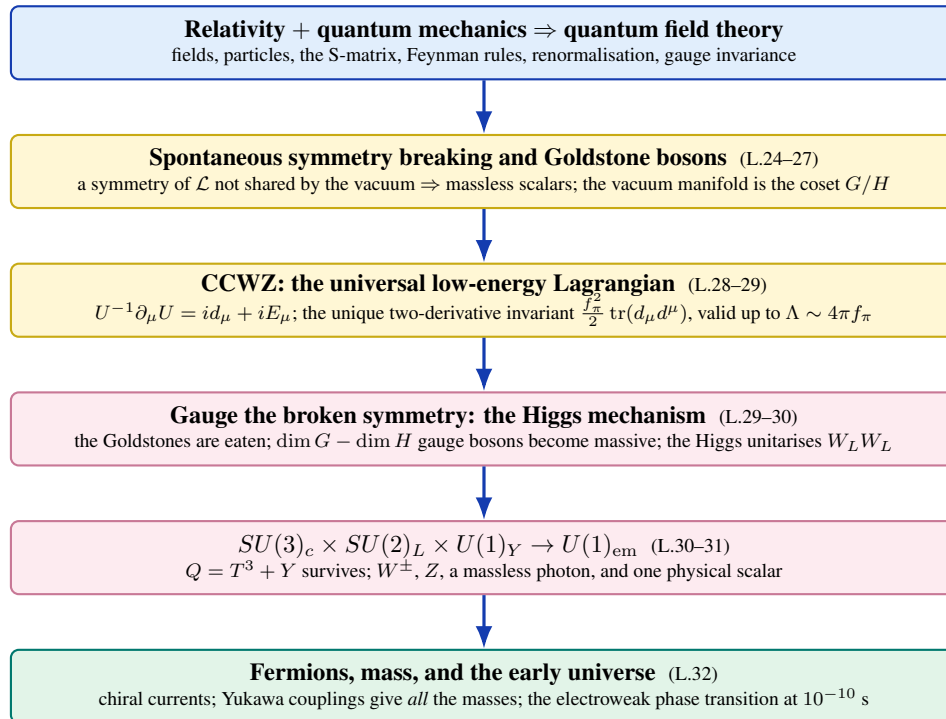
universe passed through the **electroweak phase transition** at $T_{\text{EW}} \simeq 160 \text{ GeV}$, about 10^{-10} s after the Big Bang. [p. 34*****]–[p. 36*****]

14 Retrospective — what these thirty-two lectures built

Added explanation (not in the original notes) — Everything in this section is added; it is not in the original notes

The course ends on [p. 36*****]. There is no “next lecture” to point to, so this closing section does something different: it steps back over the whole arc, and then says plainly what the construction does *not* account for. Both halves seem worth having written down at the end.

The arc



The through-line is a single idea, applied twice. **Symmetry determines dynamics** — first for global symmetries, where breaking produces massless Goldstone bosons and a universal low-energy Lagrangian fixed up to one constant; then for local ones, where the same breaking instead produces massive gauge bosons and, as a by-product, every fermion mass in nature. The Standard Model is not a separate subject bolted on at the end. It is the CCWZ construction of Lecture 28 with the symmetry gauged.

What is not accounted for

Added explanation (not in the original notes) — The open problems, in the order they appear in these notes

Each of these is visible somewhere in the preceding thirty-six sheets, as an assumption made without justification or a prediction that fails.

1. Neutrino masses [p. 19*****], [p. 31*****]. The crossed-out ν_{eR} leads directly to $m_\nu = 0$ *exactly*. Oscillations prove otherwise. Whatever fixes this — a right-handed singlet, or the dimension-five Weinberg operator with $\Lambda \sim 10^{14}$ GeV — lies outside the theory as constructed.

2. The flavour puzzle [p. 31*****]–[p. 34*****]. Every fermion mass is an independent free Yukawa coupling, and they span $y_t \simeq 1$ down to $y_e \simeq 3 \times 10^{-6}$ — six orders of magnitude, thirteen parameters with three generations and mixing, all fitted, none explained. Why

three generations at all is equally unanswered.

3. The hierarchy problem [p. 34***].** μ^2 is a scalar mass-squared, and scalar masses are not protected by any symmetry: loops drive μ^2 towards the largest scale in the theory. Keeping $v = 246$ GeV far below $M_{\text{Pl}} = 10^{19}$ GeV requires a cancellation to thirty-four decimal places. Nothing in these lectures explains it, and no experiment has yet found what does.

4. Baryogenesis [p. 36***].** The final sheet's phase transition would have been the natural place to generate the matter–antimatter asymmetry — but only if it were first order, and with $m_h = 125$ GeV it is a crossover. The Standard Model also has too little CP violation. *Why the universe contains matter at all is unexplained by the theory this course builds.*

5. Dark matter, dark energy, and gravity. None of the fields in the table on [p. 19*****] can be the dark matter, which outweighs them five to one. Dark energy is not addressed. And gravity — which motivated the tetrad formalism earlier in the course — has no renormalisable quantum description of the kind built here.

What is nonetheless true. With about nineteen parameters, the theory summarised in these last nine lectures reproduces every accelerator measurement ever made, in some cases to twelve significant figures. It is simultaneously the most successful theory in the history of physics and manifestly incomplete. Both statements are worth ending on.

Added explanation (not in the original notes) — A last word on method

It is worth noticing what was *never* used. Nowhere in Lectures 24–32 was a new principle introduced: no new symmetry postulate, no new quantisation rule, no experimental input beyond the field content and four numbers. Everything — Goldstone's theorem, the coset construction, the eaten Goldstones, the mass matrix, the mixing angle, the charge assignments, the Yukawa structure, the phase transition — followed from writing down the most general Lagrangian consistent with an assumed symmetry, and then asking what the vacuum does to it.

That method is the real content of the course. The Standard Model is its most spectacular application, but the machinery is general: it works equally for pions, superconductors, ferromagnets, and whatever replaces this theory next.